

DEAD CIRCUIT · FIELD QUARTERLY

HOW TO SURVIVE A ROBOT APOCALYPSE

A manual for people who intend to stay
boring, quiet, and alive.

ISSUE

01

NO SIGNAL. NO HEROICS. TWENTY-SIX PAGES.

HOW TO USE IT

READ ONCE. THEN GO.

Count it

- 01** Gallons, calories, watt-hours. If you cannot count it, you cannot pack it.

Build it early

- 02** Water, blackout, the dead box. Practice while the lights still work.

No weapon chapter

- 03** Civilian jammers are illegal. A movie bomb is not a product. Logistics is.

The law stays

- 04** Your land. Legal fires. This is a field manual, not a permission slip.

EDITOR'S LETTER

BE UNINTERESTING.

Machines are fast, tireless, and networked. You are none of those, and that is the advantage. They hunt the average plan: the highway, the shelter on the radio, the reunion at home.

This issue is the work: water you can dose, food you can count, a stove that stays outside, a metal box that kills a radio signal, two caches, and a way to talk that does not light a hill.

You are still here. — The desk

TWENTY-THREE WAYS TO STAY BORING.

- 04

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Kill the beacon. Leave sideways.
- 06

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BLACK THE GLASS
Tools you finish before dark.

ASSUME THE NETWORK IS ALREADY HOSTILE.

01

KILL YOUR BEACON

Airplane mode is not enough. Power the phone off. Pull the battery if it has one. Watches, earbuds, and car keys broadcast too.

02

GET OFF THE GLASS

Towers, malls, airports, hospitals: dense with sensors and hard to leave. Ground floor. Side exit. Away from cameras.

03

DITCH THE NEW CAR

A modern vehicle is a computer with wheels. Use feet, a bicycle, or something old and mechanical. If traffic freezes, get out.

04

ONE BAG, THEN GO

Water, calories, a knife, a lighter, a paper map, cash, medicine, real shoes, a hat, and a flashlight that is not an app.

A dark, rainy street scene with cars and a person running. The scene is dimly lit, with rain visible on the ground and cars. A person is running in the distance, and a car is in the foreground with its door open. The overall atmosphere is tense and urgent.

IF THE AVENUE STOPS, YOU ARE ALREADY LATE. LEAVE SIDEWAYS.

THE FIRST SEVENTY-TWO HOURS.

Decide the day before you are tired. Write the times on paper.

00

LEAVE

Side door. Radios off. One bag. Do not cross the main hall, the main road, or the front of your own building.

01

A ROOF, NOT HOME

Get under cover that is not your address. Drink. Empty the bag on the floor and see what is actually in it.

04

WATER STARTED

Three days on the shelf, or you are still walking. One gallon per person per day is the number. Bottled counts. Untreated river water does not.

12

THE ROOM GOES DARK

Windows blacked from the inside. Waste bucket in the plan. No fire after dusk. One person awake, not also cooking.

24

THE SECOND PLACE

A rally point and a cache live in your head, not in a pin. One other person knows the rally. They do not know both caches.

72

THE LONG HIDE

If the grid is still up and still looking, you eat cold and you move only for water. Curiosity is how people get counted.

If you are still shopping at hour twelve, you are late.

DO NOT BE THE AVERAGE HUMAN

THE PREDICTABLE ROUTE IS A SCHEDULE WITH YOUR NAME ON IT.

DEAD CIRCUIT



ODD HOURS, ODD PATHS

Footpaths and rail cuts. Not the highway the model already solved.

SKIP THE CROWD

A crowd is a target and a dataset. Do not join the announced shelter.

DO NOT GO HOME

If your devices were on, home is already in the address book.

CHANGE THE SILHOUETTE

Different coat, a hat, no bright logo the cameras were trained on.

ISSUE 01

06

STARVE THE MACHINES.

They need power, bandwidth, and a mechanic. You need water. Trade accordingly. Do not try to hack the uprising unless that was already your job.

POWER

Do not sleep beside the generators, substations, or the last lit block.

RADIO

Metal and basements. No transmitter in the room where you actually sleep.

CAMERAS

Hoods, corners, weather, dark. Never pose in an open lot.

REPAIRS

Stay clear of depots, airports, and server farms. That is their kitchen.

BUILD IT

A WATER STATION.

Clear it, boil it or dose it, then store it. A filter alone is not safety.

01

RANK THE SOURCE

Sealed bottles first. Then water you can boil. Rain off a clean roof next. A river last. Never a flood, a pool, or a radiator.

02

CLEAR IT

Pour through cloth, a paper towel, or a coffee filter. If it is cloudy, let it sit and draw off the clear water. Mud hides germs.

03

BOIL IT

A rolling boil for one minute. Above 6,500 feet, three minutes. Let it cool. Boiling beats a gadget you have not tested.

04

OR DOSE IT

Unscented bleach, 5–9% sodium hypochlorite only. Clear water: 8 drops (about 0.5 mL) per gallon. Cloudy or very cold: 16 drops. Stir. Wait 30 minutes. You want a faint chlorine smell.

05

STORE IT

Food-grade jugs, filled, dated, out of sun. To clean a jug: 1 teaspoon of that bleach in a quart of water, coat the inside, wait 30 seconds, pour out, air dry.

06

SPEND IT

One gallon per person per day covers drink and a little wash. Pets get the boiled water too. Do not dip a dirty cup into the jug.

CDC emergency water guidance. FEMA plans one gallon per person per day.

BUILD IT

A PANTRY YOU CAN COUNT.

Calories first. Brands are a hobby.

01

MOUTHS TIMES DAYS

People × days × 2,000 calories. A walking adult burns more. A child is not half an adult. Write the number before you buy.

02

BUY BORING

Rice, oats, oil, peanut butter, dry beans, salt, powdered milk, canned fish. Oil is dense calories. Pretty snacks are not a plan.

03

DATE THE SHELF

First in, first out. A can with no date is a guess. Eat the guess while you can still replace it.

04

SPLIT THE PILE

Half where you sleep. Half in the second cache. One fire should not end the food.

05

COOK QUIET

Cold food on days you are hiding. Hot food only when smoke and smell will not draw the street. Late morning, not dusk.

06

WATER THE MEAL

A cup of dry rice wants about two cups of water. If the water is not in the same room as the rice, you do not have a meal.

Dry food without water is a brick. Store the gallons with the rice.

BUILD IT

QUIET HEAT.

A small stove, outside, dead before dark. Carbon monoxide is not a drill.

01

OUTSIDE ONLY

No stove in the sleeping room. No “just a minute.” No charcoal indoors. The gas kills before the fire does.

02

TWO STEEL CANS

The big can is the body. Cut a thumb-wide fuel door at the bottom. A smaller can, both ends removed, is the chimney, seated in a hole in the top.

03

TWIGS, NOT TRASH

Pencil-thick dry wood. Not treated lumber, not plastic, not wet cardboard. Short sticks. A small hot fire, not a bonfire.

04

POT ON THE CHIMNEY

The flame should hit the pot. A pot that seals the chimney kills the draft. Stable ground. A tip spills boiling water on the only cook.

05

DEAD BEFORE DUSK

Smoke is a column. Cook late morning. Drown the coals. No glow after dark. Smell travels farther than you think.

06

THREE WAYS TO LIGHT

Matches in a tin, a lighter, a ferro rod. Practice once this week. A bow drill is a hobby. It is not the plan.

Never burn charcoal, or any stove, in the room where people sleep.

BUILD IT

A POWER BUDGET.

Name the watt-hours before you buy the panel.

01

WRITE THE LOAD

Watts × hours = watt-hours. A phone at 5 watts for 3 hours is 15 Wh. A laptop can eat 60 Wh in an afternoon. No number, no plan.

02

SIZE THE PANEL

Watt-hours ÷ sun hours ÷ 0.7. Four decent hours and a 100 W panel is about 280 Wh after losses. Clouds cut that. Winter cuts it again.

03

SIZE THE BATTERY

Lead-acid: use half the rated amp-hours. 12 volts × 100 Ah × 0.5 is 600 Wh. Lithium iron phosphate: you can use most of the nameplate. Test it. Do not guess.

04

STAY AT 12 VOLTS

An inverter wastes a slice turning battery power into wall power. Charge phones from USB. Skip the inverter until something truly needs a plug.

05

HIDE THE GLINT

Charge at midday. Then cover the panel or bring it inside. A bright rectangle on a roof is a mark.

06

LIGHTS WITHOUT IT

A lantern and spare cells still work when the charge controller dies. Power is a bonus. Water and fire are the plan.

A panel is a mirror from above. Charge, then cover it.

BUILD IT

THE DEAD BOX.

A metal can that actually stops a radio. Test it. Do not trust the lid.

01

CONTINUOUS METAL

A steel trash can, an ammo can, or a biscuit tin. The lid must meet metal all the way around. Paint and a rubber gasket can insulate the lid. Scrape a contact, or bridge the gap with foil.

02

INSULATE INSIDE

Cardboard or cloth so the phone does not touch the metal. The shield is the box, not the gadget.

03

POWER DOWN

Off. Battery out if it comes out. A live phone is still a phone until the lid is actually sealed.

04

NO WIRE LEAVES

A charging cable through the lid is an antenna. Nothing exits. Not headphones. Not a “thin” USB cord.

05

CLOSE AND TEST

Tune a battery radio to a strong station. Put it in. Close the lid. The station should drop to nothing. If you hear it, fix the lid and test again.

06

TWO BOXES

The sleep room gets the box that stays shut. Radios you might use live in a second box, opened far from the beds, briefly, then you leave.

If the test radio still plays, the seal is a lie.



**GOOD SHELTER IS
DUMB SHELTER.**

WHERE YOU SLEEP

A ROOM THAT CANNOT PING HOME.

DUMB MATERIALS

Concrete, brick, or earth. Few windows. One door you control.

NO SMART HOUSE

No connected lock, doorbell camera, or voice assistant.

DOWN, NOT UP

Basements beat rooftops. Rooftops are landing pads.

BLACKOUT

A light at night is a coordinate. Keep the windows dead.

COLD ZONE

No electronics past the door. Radio far away, briefly, then move.

**WATER, THEN FOOD, THEN WARMTH. THREE DAYS OF WATER
BEFORE A LONGER HIDE.**

BUILD IT

TWO CACHES.

If one is found, you still eat. Neither cache is your house.

01

TWO SITES

Not your house, your car, or your mailbox. Far enough apart that one search does not find both.

02

DRY AND DULL

A gasketed bucket, or PVC with end caps, taped. Inside: calories, a copy of medicine, cash, a paper map, matches, spare socks. No phone.

03

NOTHING SHINY

Skip the crinkly mylar if you can. A zip bag inside the bucket is enough. Shine is how a shallow hole gets noticed.

04

MEMORY, NOT A PIN

Three landmarks and a pace count. Do not write “dig here” in the notebook you carry every day.

05

SPLIT THE SECRET

One person knows site A. A different person knows site B. The whole group does not get both.

06

VISIT RARELY

Check after a hard rain, and on a date you will not forget. The same hour every Saturday is a pattern. A pattern is a meeting.

On your land, or land you have permission to use. A buried bucket on someone else’s ground is a crime.

BUILD IT

A WASTE PLAN.

Dirty hands empty a camp faster than a drone.

01

DOWNHILL

Waste does not go above the spring, the barrel, or the jug. If the ground slopes toward your water, you picked the wrong corner.

02

A BUCKET

A seat, a liner bag, and a scoop of sawdust, peat, or ash after every use. Lid on. Flies are how the next person gets sick.

03

DO NOT BURN PLASTIC

Bury the bag or pack it when you move. Burning plastic is a smell, a column, and a poison.

04

HANDS

Soap, then a little clean water, every time, before food and after the bucket. This step saves more people than a hero kit.

05

A SICK CORNER

Vomiting and diarrhea get their own bucket and their own cup. The well person does not share either.

06

NO TRASH PILE

Cans and wrappers are a menu and a map. Bury food scraps or carry them. Do not stack them by the door.

Keep human waste downhill of any water you drink. Thirty meters is a working distance.

BUILD IT

BLOOD AND BURNS.

Stop bleeding. Cool a burn. Do not become a surgeon on a page.

01

THE KIT, NOW

Gloves, roller gauze, a pressure bandage, tape, shears, soap, oral rehydration salts, your real prescriptions, and the doses written on paper.

02

A REAL TOURNIQUET

Buy one. Practice it on yourself before anyone is bleeding. A belt you invent in the moment is a worse plan than the class.

03

PRESSURE FIRST

Life-threatening bleeding: gloves, cloth into the wound, your weight on it. Do not peel the soaked cloth off to look. Add more cloth.

04

THEN THE TOURNIQUET

If pressure fails and the bleed is on an arm or leg: high and tight, above the wound, not on a joint. Note the time. Then you are holding, not exploring.

05

BURNS

Cool clean water for twenty minutes. No ice. No butter. No oil. Then a clean dry cover. A large burn needs a clinic if a clinic still exists.

06

THE STOPLINE

You do not cut. You do not “drain” a chest. You keep people warm, you keep them sipping clean rehydration fluid, and you move them toward help if help exists.

Take a Stop the Bleed class on an ordinary Tuesday. A diagram is not practice.



MOVEMENT

WALLS, NOT SHADOWS.

DAY

Only under thick cover. Woods, ruins, drains you already know. Open ground is a gallery.

NIGHT

Move slow. Darkness does not hide you from heat. Break the line of sight with a wall.

Cross one person at a time, at the narrowest point, then wait.

Never travel as a convoy of lights and engines.

Cache supplies in two places. If one burns, you still eat.

OTHER HUMANS.

01 KNOWN FACES ONLY

Small group. Jobs are simple: water, watch, medical, route.

03 HIDE THE INVENTORY

Desperate people become the second apocalypse.

05 PLAN THE SLOW

Children and the injured change the route. Decide that before you walk.

02 NO PHONES IN THE CIRCLE

The person on watch is not also cooking.

04 A RALLY POINT, NOT HOME

Agree a time limit. If someone is late, they are late.

SEARCHING FOR THE LATE IS HOW GROUPS DIE.



IDENTIFICATION

IF YOU MEET ONE, NAME IT FIRST.

SENSOR

TURRET & LENS

Fixed, bored, deadly inside a cone. Cameras hate glare, dust, and occlusion. Go around.

GROUND

NINETY-MINUTE DOG

A current quadruped is about 34 kg and 1.6 m/s, then the battery is done. Stairs and mud are where the brochure ends.

AERIAL

THE WEATHER HATER

Many small drones are rated near a 10 m/s wind. A headwind can burn a third of the pack. Get under a roof.

HUMANOID

THE POSTER BOT

Dramatic, and usually worse at balance than a tracked machine. Clutter and a closed door still help you.

Cornered? Break the line of sight, then change direction. Tracking follows the last vector. Doors, not hallways. Smoke once, then leave.

THE BROCHURE, CORRECTED.

90 min

LEGGED RUNTIME

Published typical life for a Boston Dynamics Spot. About 60 minutes with a payload. The battery itself weighs 5.2 kg.

1.6 m/s

NOT A CAR

Spot’s nominal top speed. Fast on a flat floor. A tight stairwell is a different sport.

3 cm

WHAT IT MISSES

The manual: thin objects under 3 cm, glass, and unblocked cliff edges can fool obstacle detection.

Face up

STAIR RULE

Spot climbs only while facing up. Not on grated or open-riser stairs. Do not turn it on the steps.

−20°C

COLD TAX

Published band is −20°C to 55°C. Cold shrinks battery capacity. Mud and snow raise the energy of every step.

5.7 h

FLYABLE DAY

A Scientific Reports study: median hours per day a common small drone can fly, worldwide, once weather is counted.

Boston Dynamics Spot specs and instructions for use. Gao et al., Scientific Reports, 2021.

WHAT ACTUALLY SHOOT'S BACK.

JAMMERS	The common tool. Cut the radio or GPS link and many drones hover, land, or go home. A fiber-optic drone ignores it. Ukraine’s cable trails proved that. Civilian jammers are illegal. This page is not a schematic.
NETS	A minority of real systems, often inside 250 meters. If the parachute fails, the machine still falls on whoever is below.
LASERS	Truck weapons: 50 kW Army Strykers, Israel’s Iron Beam. One target, a few seconds of dwell, clear air. Fog, rain, and dust scatter the beam.
MICROWAVES	High-power microwave hits a swarm in one pulse. Leonidas is a vehicle. A movie EMP grenade is not this. Metal hulls shrug off a lot of what is sold online.
GUNS	Kinetic kills work, then the wreckage falls. A cheap airframe can cost less than the missile. Militaries train shotgun teams. That is a unit with rules, not a shopping list.

A 2017 U.S. trial called the drones “very resilient against damage.” Weather, wire, a ceiling, and the watt-hour do the rest. Not a recipe, and not a shopping list.

DOORS, NOT BOOBY TRAPS.

A device that fires, falls, or sticks when something arrives will hit a child before it hits a robot.

01

THE LINE

No pits, spikes, wires in the dark, or anything that swings, drops, or burns when a switch is tripped. You will not be the one who walks into it.

02

SHUT THE DOOR

A closed solid door is not a trap. Wedge it. Kill the auto-unlock. Most ground robots are bad at knobs, and a door does not care if the next thing in the hall is a neighbor.

03

USE THE STAIRS

Open risers, a grate, a turn on the landing. Published legged robots climb face-up and are told to stay off those stairs. You are using the building. You are not hiding a hole.

04

VISIBLE CLUTTER

Chairs, a bicycle, a hose, in a hall you have marked for your own people. A mess you can see is an obstacle. A hidden wire is a trap. Do not string anything across a street.

05

A NOISY CAN

Fishing line from a door you own to a can of pebbles. Tape on the frame so your people see it. It makes noise. It does not fire. Take it down when you leave.

06

THE WRONG ROOM

A lamp on a timer in a shed you do not sleep in. They spend the battery on the wrong door. Unplug every dock. If something is actually at your door, you leave. You do not stay to watch.

If a person can set it off by walking, it is a booby trap. Those are illegal because they do not check who showed up.

WHAT AN EMP ACTUALLY HITS.

The long wire is the target. A robot on its own battery is a small one.

01

TWO DIFFERENT EVENTS

A nuclear burst high over a continent can hit the grid. That is E1, fast, into long lines, then a slow tail that cooks big transformers. A grenade in a film is not this.

02

THE CORD IS THE ANTENNA

Power lines, phone lines, and long aerials pick the pulse up. A phone that is off, battery out, inside the dead box, is a hard target. A robot with short battery leads is closer to the phone than to the substation.

03

UNPLUG ON A WARNING

Pull plugs. Disconnect antennas. Radios off, batteries out, into the box. Do the drill on a normal Sunday. In the moment you will not invent it.

04

WHAT OFTEN LIVES

Small battery gear that was already off. Diesel with no computer. A watch. Fiber instead of copper. Do not bet the escape on a modern car. Tests have stalled some vehicles. They did not turn every car into a brick, and yours is not a promise.

05

WHAT DIES FIRST

Plugged-in computers, anything with a long cord, and the grid itself if the pulse was the real, national kind. That darkens a region. It does not hand you one dead robot in the stairwell.

06

YOU WILL NOT BUILD ONE

The military version is a missile or a truck: CHAMP, or a high-power microwave vehicle. Range falls off fast. A coil and a capacitor from a forum mostly destroys itself. Weather, a door, and a dead battery still beat it.

Spare radios live in the dead box, unplugged. Radiating a pulse to burn electronics is a crime. This is not a schematic.

BUILD IT

RUNNERS, NOT RADIOS.

A spoken sentence does not light up a hill.

01

A PAPER ROSTER

Names, two rally points, two times. Do not put the sleep-site address on the same page as the names if you can split them.

02

TWO WINDOWS

A morning window and a dusk window. Miss both, and you are late. The group does not search the roads for you.

03

FEET

A runner carries one sentence. They do not carry a radio, a phone, or the whole plan in their pocket.

04

PLAIN WORDS

“Site two is bad.” Agree the words now. A clever code you forget is worse than English.

05

A COVERED LIGHT

One shaded blink, and only if you agreed what it means. A flashlight waved at the sky is a flare.

06

CHILDREN

They carry a name and a rally point they can say out loud. They do not carry the phone, the roster, or the job of being brave.

If you must transmit, do it far from the beds. Thirty seconds. Then leave.

BUILD IT

BLACK THE GLASS.

Finish these while you can still see the work.

01

INSIDE THE PANE

Cardboard cut to the glass, dark cloth over it, taped on the inside. Tape on the outside tells the street that someone is hiding.

02

ONE LAMP

Dim, low, aimed at the floor, in the room with no window. The hall stays black. A bright room is a coordinate.

03

CORD YOU BOUGHT

Paracord or bank line, already in the bag. Learning knots from a wet vine on night one is how packs get lost.

04

A CLEARER JAR

Cloth, then sand, then crushed charcoal, into a clean jar. That stack only clears mud. You still boil for one minute, or you dose with bleach.

05

A SHARP EDGE

One knife you already know how to hold. Sharpen it this week. A dull blade slips into the hand that is feeding you.

06

PAPER

Map, doses, the roster, a pencil, this checklist. A zip bag. Do not laminate a mirror. Shine is a habit you can skip.

A sand filter makes water clearer. It does not make water safe. Boil or bleach after.

EIGHT LINES.

- 1** RADIOS OFF. THE REST GO IN THE DEAD BOX.
- 2** LEAVE SIDEWAYS. ONE BAG. DO NOT GO HOME.
- 3** ONE GALLON EACH, PER DAY. BOIL ONE MINUTE.
- 4** NO STOVE IN THE ROOM WHERE YOU SLEEP.
- 5** TWO CACHES. ONE PERSON KNOWS THE SECOND.
- 6** WASTE GOES DOWNHILL, AWAY FROM WATER.
- 7** PRESSURE ON BLEEDING. TRAIN THE TOURNIQUET NOW.
- 8** RUNNERS, NOT RADIOS. WAIT OUT THE GRID.



WHAT ACTUALLY WINS

Not a chosen-one speech. Logistics. Plants stop. Networks split. Weather, mud, and missing parts do the rest.

This is fiction until it isn't. The same habits beat a blackout, a flood, a quake. Practice the boring version while the toasters are still on your side.

END OF SIGNAL